

Lab 1: Decision Trees

DD2421 Machine Learning

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Assignment 0: Dataset Difficulty

Question: Which dataset is the most difficult to learn?

Monk-1: $(a_1 = a_2) \vee (a_5 = 1)$

This set is the easiest. Only 3 attributes are needed to be known to predict (i.e optimal tree depth is 3).

Monk-2: a_i for exactly 2 i

Decision trees struggle with this because the information is not contained in individual attribute splits. Each attribute by itself contains zero information. We need to know all 5 attributes simultaneously to classify a sample (i.e tree depth is 5).

Monk-3 $(a_5 = 1 \wedge a_4 = 1) \vee (a_5 \neq 4 \wedge a_4 \neq 3)$

This set is the second hardest. It has 5% noise which will lead to bad generalization. But it won't need much more than a depth of 3.

Assignment 1 & 2: Entropy

Assignment 1: Dataset Entropy

- Monk-1: 1.0
- Monk-2: 0.957
- Monk-3: 0.999

Assignment 2: Understanding Entropy

- **Definition:** Entropy quantifies the uncertainty in a distribution.
- **Uniform Distribution (Max Entropy):** All outcomes are equally likely and is hardest to predict. Example: A fair coin
- **Skewed Distribution (Low Entropy):** One outcome dominates and is easier to predict. Example: Height measurement

Assignment 3: Information Gain

Selecting the root node: We select the attribute with the highest Information Gain (IG).

Dataset	A1	A2	A3	A4	A5	A6
Monk-1	0.075	0.006	0.005	0.026	0.287	0.001
Monk-2	0.004	0.001	0.001	0.016	0.017	0.006
Monk-3	0.007	0.294	0.001	0.003	0.256	0.007

- **Monk-1:** Split on A5.
- **Monk-2:** Split on A5.
- **Monk-3:** Split on A2.

Assignment 4: Motivation for Information Gain

Why maximize Information Gain?

$$\text{Gain}(S, A) = H[S] - \sum_{v \in \text{value}(A)} \frac{|S_{A=v}|}{|S|} H[S|A=v] = H[S] - E[H[S|A]]$$

- **Reducing Impurity:** Maximizing IG is mathematically equivalent to minimizing the expected entropy of a attribute A. We choose this as the root then repeat for its children. We want subsets that are as pure as possible
- **Heuristic Efficiency:** By choosing the attribute that best separates the classes early on, we create a shallower, simpler tree.
- **Generalization:** Simpler trees tend to generalize better to unseen data compared to unnecessarily deep trees

Assignment 5: Full Tree Performance

We built full decision trees using ID3 and evaluated them on the test sets.

Dataset	E_{train}	E_{test}
Monk-1	0.0	0.1713
Monk-2	0.0	0.3079
Monk-3	0.0	0.0556

Observations:

- E_{train} is 0 for all, indicating the trees fully memorized the training data (overfitting).
- Monk-2 has the highest test error, confirming it is the hardest concept to learn using standard splits

Assignment 6: Pruning & Bias-Variance

The Bias-Variance Trade-off

- **Full Trees (High Variance):** They fit the training data perfectly (low bias) but capture noise, leading to poor generalization (high variance).
- **Pruning (Reducing Variance):** By removing branches that don't provide much information, we simplify the model.
- Pruning slightly increases bias (we might miss some training nuances) but significantly decreases variance so the model becomes less sensitive to the training data set. leading to a lower total error on the test set

Assignment 7: Pruning Analysis

```
1: function PRUNETREE( $T, V$ )
2:    $T_{current} \leftarrow T$ 
3:    $bestAcc \leftarrow \text{ACCURACY}(T_{current}, V)$ 
4:   while true do
5:      $Candidates \leftarrow \text{ALLPRUNED}(T_{current})$ 
6:     if  $Candidates$  is empty then
7:       return  $T_{current}$ 
8:     end if
9:      $T_{best} \leftarrow \arg \max_{T' \in Candidates} \text{ACCURACY}(T', V)$ 
10:     $candidateAcc \leftarrow \text{ACCURACY}(T_{best}, V)$ 
11:    if  $candidateAcc \geq bestAcc$  then
12:       $T_{current} \leftarrow T_{best}$ 
13:       $bestAcc \leftarrow candidateAcc$ 
14:    else
15:      return  $T_{current}$ 
16:    end if
17:  end while
```

Assignment 7: Pruning Analysis

We performed Reduced Error Pruning with different training/validation splits ($N = 5000$ iterations where the split was shuffled).

- **Monk-1:** Optimal performance around split 60:40 but has slightly higher variance. Less biased for less training data.
- **Monk-3:** Optimal performance around split 70:30 and slightly lower variance. Higher variance for less training data due to noise

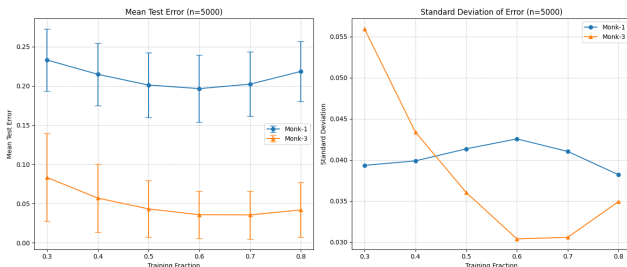
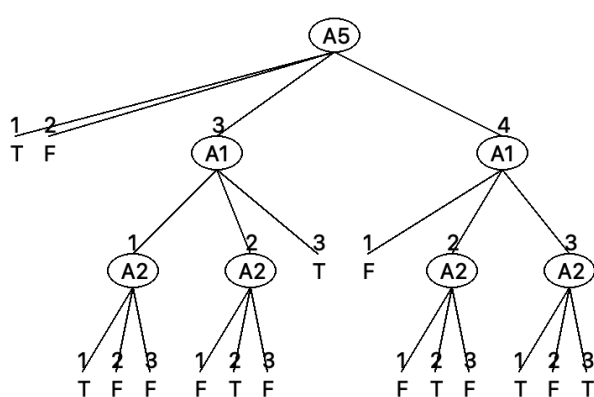


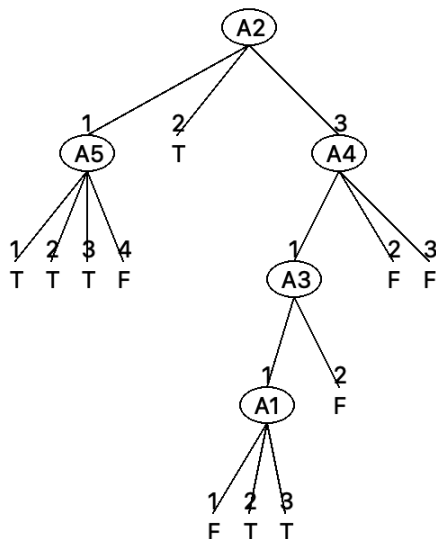
Figure: Mean test error vs. Training Fraction

Assignment 7: Pruning Analysis



Monk 1

Assignment 7: Pruning Analysis



Monk 3